International Journal of Sports Physiology and Performance, 2022, 17, 569-575
https://doi.org/10.1123/ijspp.2021-0292
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ORIGINAL INVESTIGATION

Sleep Architecture in Response to a Late Evening Competition in Team-Sport Athletes

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Purpose: The aim of this study was to assess the effect of a late evening competition on sleep characteristics and perceived recovery in team-sport players using a validated self-applicable and portable polysomnographic device. ***Methods:*** Sixteen team-sport players (age: 25.4 [1.4] y; body mass index: 23.6 [0.5] kg/m^2) completed the study. Objective and subjective sleep data were collected for 4 consecutive nights: 2 nights before (PRE2 and PRE1) and the 2 nights after an evening match (POST1), performed between 6:00 PM and 9:00 PM. Total quality of recovery values were also collected in the morning. ***Results:*** A significantly delayed bedtime ($P < .0001$; $\eta_p^2 = .68$, large) was observed in the first night after the competition (2 h 29 min [1 h 15 min]) compared both to PRE2 (+88 min; $P < .0001$), PRE1 (+98 min; $P < .0001$), and POST1 (+100 min; $P < .0001$), and similar results were observed for wake-up time ($P = .033$; $\eta_p^2 = .39$, large): Players woke up significantly later in evening match (9 h 20 min [1 h 55 min]) compared with PRE2 (+85 min; $P = .050$) and POST1 (+85 min; $P = .049$). Conversely, total sleep time; sleep efficiency; sleep onset latency; wake after sleep onset; cortical arousals; N1, N2, N3, and REM (rapid eye movement) percentages; total quality of recovery values; and scores of subjective sleep quality did not vary among the 4 study nights. ***Conclusions:*** Team-sport players had delayed bedtime and wake-up time following an evening competition; however, sleep quality, duration, and subjective scores of recovery were not affected by the evening match. The delayed wake-up time seems to protect athletes' sleep efficiency/duration against the evening-match-induced delayed bedtime.

Keywords: REM, sleep quality, time of day, evening match

Elite athletes often do not reach the recommended night-time sleep duration of 7 to 9 h/night[1] and acute sleep restriction could negatively impact both sport-specific and athletic performance in different sport disciplines.[2–4] Sleep disturbances in athletes may be explained by their constant exposure to many stressors that impair sleep, such as high training loads and high-intensity training sessions, long-haul travels and high levels of precompetition anxiety and stress.[5,6] Among these variables, also training and competition schedules may influence the sleep response to exercise.[7] For this reason, the American Sleep Association suggests avoiding high-intensity exercise in the evening as an effective sleep hygiene strategy as it could potentially increase arousal, core body temperature, and cortisol production upon bedtime and affect subsequent sleep.[8] Nonetheless, the negative effect of late evening high-intensity exercise on sleep has not always been observed. Recent experimental studies highlighted that moderate- to high-intensity training did not disrupt nocturnal sleep in nonathlete and athlete populations,[9–11] and a systematic review by Stutz et al[12] concluded that exercising in the evening does not negatively affect sleep in healthy participants.

Only few studies investigated sleep changes in team sport athletes when the physical demand is at the highest levels during the evening, namely following a night competition or match. Nédélec et al[13] studied the sleep of Champions League soccer players after night matches and observed that total sleep time was reduced when compared with training days, and this difference was explained by the delayed bedtime after matches. Similarly, Vitale et al[14] detected lower actigraphy-based sleep quality and total sleep time in elite volleyball players the first night after an evening match compared with prematch sleep. Therefore, as also confirmed by a recent meta-analysis by Roberts et al,[5] it seems that total sleep time and sleep efficiency (SE) are reduced during the night of evening competition compared with precompetition nights or training periods.

Interestingly, all the above-mentioned studies investigated sleep in athletes only through wrist actigraphy. Although wrist actigraphy offers many advantages when compared with polysomnography (PSG; eg, less invasive and sleep monitoring for many consecutive weeks), its main limitation concerns the impossibility to record the biological signals crucial for sleep staging: electroencephalography, electrooculography, and electromyography.[15] On the contrary, PSG is ideal for sleep staging and the evaluation of neurophysiological features, but it is limited by high costs for recording and scoring, and short evaluation periods.[16] Importantly, new portable devices that bridge the gap between laboratory PSG and actigraphy provide a striking alternative. Indeed, devices embedding the 3 above-mentioned fundamental signals for sleep staging allow both the identification of sleep waves and multiple nights' recording.[17–19]

To the best of our knowledge, no study yet examined the impact of an evening competition on team sport athletes' sleep architecture utilizing a validated portable polysomnographic device, mixing the comfort of actigraphy and the specific information provided by PSG. Therefore, the primary aim of this study was to assess the effect of a late evening competition on objective and subjective sleep parameters and perceived recovery in team sport players. Based on the existing literature,[5] we hypothesized to detect

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a delayed bedtime and a decrease in sleep quality and duration following the evening competition.

Materials and Methods

Study Design

This was an observational study conducted between January and May 2019 at the IRCCS Istituto Ortopedico Galeazzi (Milan, Italy) in accordance with the Strengthening the Reporting of Observational studies in Epidemiology guidelines.[20] The study was approved by the Ethical Committee of Vita-Salute San Raffaele University (ref. no.: 134/int/2018), and all procedures were performed in compliance with laws and regulations governing the use of human subjects (Declaration of Helsinki). The study protocol was registered at clinicaltrials.gov (ClinicalTrials.gov Identifier: NCT03773601). All participants received explanation of purpose, methods of the study, and written informed consent was obtained from all subjects.

Subjects

The participants were 16 Italian athletes (age: 25.4 [1.4] y; height: 1.85 [0.02] m; weight: 81.0 [2.5] kg; body mass index: 23.6 [0.5] kg/m^2), practicing field hockey (n = 8), volleyball (n = 4), and basketball (n = 4) at regional and national level with a mean weekly training volume of 15.6 (1.4) hours. The subjects were recruited upon invitation by their coaches. Participants were screened for eligibility using the following inclusion criteria: male gender, all ethnicity, age between 18 and 45 years, practicing a team sport ≥8 hours per week for at least 6 years. The exclusion criteria were shift- or part-time workers, tobacco use, history of injury affecting the frequency of training, and use of medications able to influence sleep. Participants who met the above inclusion and exclusion criteria were recruited in the study. Sleep data were collected for 4 consecutive nights and analyzed as 2 nights before (PRE2 and PRE1 separately) and the 2 nights after an evening match (EM and POST1 separately), starting between 18:00 and 21:00, and with a finish time between 20:00 and 23:00.

Chronotype and Baseline Subjective Sleep Measurements

Circadian typology (ie, the chronotype) was assessed via the Horne–Ostberg Morningness-Eveningness Questionnaire.[21] Participants were classified as morning type (score ≥ 59) or evening type (score ≤ 41) or neither type (score 42–58). The Pittsburgh Sleep Quality Index was used to evaluate subjects' subjective sleep quality and to assess possible sleep complaints.[22]

Subjective Recovery Scores

The Total Quality Recovery (TQR) scale measures the perceived recovery process in sport.[23] It is a verbal-anchored scale, with a score ranging from 6 (very poor recovery) to 20 (very good recovery), providing a means to measure the athlete's psychophysiological recovery. TQR values were collected in the morning at PRE2, PRE1, EM, and POST1.

Sleep Architecture

All players wore the Sleep Profiler (Advanced Brain Monitoring, Carlsbad, CA) to record their sleep parameters for 4 consecutive nights (PRE2, PRE1, EM, and POST1). The Sleep Profiler is a multichannel forehead recorder with frontopolar derivations (AF7, Fpz, AF8) providing the electroencephalography, electrooculography, and electromyography signals requested for sleep staging. Electroencephalographic signal was sampled at 256 Hz and filtered with a 0.1-Hz high-pass and 80-Hz low-pass filter, and notch filtered at 50 Hz. Sleep scoring was automatically performed for 30-second epochs throughout a dedicated software and a procedure that have been previously validated. In detail, a strong agreement was observed between the automated sleep staging and human-scored PSG: The mean overall interscorer agreements between the technologists were 75.9%, the mean kappa score was .70, and the authors concluded that the automated analysis was as accurate as the human-staged PSG.[17]

Each participant completed a daily sleep diary to record bedtime, wake-up time, and the number and duration of nocturnal awakenings. Data derived both from the sleep device and sleep diaries were used to determine the quantity and quality of sleep that participants obtained. In detail, the following 11 parameters have been analyzed:

1. Bedtime: the time at which a participant attempted to sleep in bed
2. Wake-up time: time of the final wake-up
3. Total sleep time: recording time from the bedtime minus the time spent awake
4. SE: total sleep time divided by time in bed, expressed in percentage
5. % rapid-eye-movement (REM) sleep: hours of valid REM sleep divided by hours of sleep time
6. % stage N1: hours of valid stage 1 sleep divided by hours of sleep time
7. % stage N2: hours of valid stage 2 sleep divided by hours of sleep time
8. % stage N3: hours of valid stage 3 sleep divided by hours of sleep time
9. Sleep onset latency: elapsed time from the attempt to sleep until the start of the first 3 consecutive nonwake epochs at the start of the night
10. Wake after sleep onset (WASO): all minutes the subject was awake after sleep onset until the end of the record
11. Cortical arousals: number of cortical arousals occurrences per hour of sleep time

Furthermore, the sleep diary served to record subjective sleep quality (from 0 = very poor sleep quality to 10 = optimal sleep quality).

Statistical Analysis

Power Analysis. To determine the sample size of the study, *a priori* power analysis based on the existing scientific literature[5] was performed. The main outcome was to determine if there was a difference in SE following a late evening competition. With the following parameters, $\alpha = .05$, statistical power = 90%, and effect size = 1.0, it was calculated that a sample of 13 participants would be enough to detect a significant change in SE after a night competition. To be more conservative and to prevent a possible drop of participants (estimated in 15%–20%), the total sample was increased to 16 athletes. To calculate the sample size, the G*Power software (Universität Düsseldorf, Düsseldorf, Germany) was used.

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Effect of the Late Evening Competition. Descriptive statistics (mean [SD]) for the outcome measures were calculated. Each sleep parameter and TQR score was calculated 4 times (PRE2, PRE1, EM, and POST1) for all subjects and then checked for normality with the Shapiro–Wilk test. All parameters showed normal distribution with the exception of SE, N1, sleep onset latency, WASO, cortical arousals, and subjective sleep quality. To detect possible differences in sleep among PRE2, PRE1, EM, POST1, a repeated-measures analysis of variance followed by the Bonferroni post hoc test, or the equivalent nonparametric Friedman test followed by the Dunn's multiple comparisons for nonnormally distributed variables, was performed. Partial eta-squared (η_p^2) was used to determine the magnitude of the effect for significant outcomes ($\alpha = .05$) in the analysis of variance using the small (<.13), medium (.13–.25), and large (>.25) interpretation for effect size.[24] A P value ≤.05 was considered statistically significant. Statistical analysis was performed using GraphPad Prism (version 6.00; GraphPad Software, San Diego, CA).

Results

Twenty-five participants were screened for eligibility and met the inclusion criteria. A total of 9 athletes dropped out: 4 due to the nonadherence to the experimental procedures (ie, not wearing the device for 4 consecutive nights), 2 because the sleep diary was not correctly filled, and 3 for technical problem during data acquisition. Therefore, 16 players completed the study and were classified as 14 neither types, 1 morning type, and 1 evening type. At baseline, Pittsburgh Sleep Quality Index mean score was 4.2 (0.3).

Sleep Parameters

Figure 1 shows the box and whisker plots of the objective and subjective sleep parameters at PRE2, PRE1, EM, POST1, whereas Table 1 displays mean (SD) of sleep parameters and TQR values, the results of repeated-measures analysis of variance procedure and η_p^2 for significant outcomes.

A significantly delayed bedtime ($P < .0001$; $\eta_p^2 = .68$, large) was observed in the first night after the competition (2 h 29 min [1 h 15 min]) compared both to PRE2 (+88 min; $P < .0001$), PRE1 (+98 min; $P < .0001$), and POST1 (+100 min; $P < .0001$). Similarly, repeated-measures analysis of variance highlighted significant differences in wake-up time among the 4 study nights ($P = .033$; $\eta_p^2 = .39$, large): Players woke up significantly later after the evening match (9 h 20 min [1 h 55 min]) compared with PRE2 (+85 min; $P = .050$) and POST1 (+85 min; $P = .049$). On the contrary, no other sleep parameter has significantly been changed in relation to the late evening competition.

Subjective Recovery

In line with the latter result, TQR values and scores of subjective sleep quality were similar among PRE2, PRE1, EM, and POST2.

Discussion

To our knowledge, this is the first study that investigated the effect of an evening competition on athletes' sleep architecture by the use of a validated portable polysomnographic device. In detail, we observed a delayed bedtime and wake-up time in the first night following the competition. Strikingly, sleep architecture, TQR values, and subjective sleep quality did not display any significant changes among the study nights. Our initial hypotheses were partially confirmed.

In our study, team sport athletes obtained the National Sleep Foundation recommendations for SE (ie, >85%), sleep onset latency (ie, <30 min), and WASO (ie, <40 min)[25]; however, they did not attain the National Sleep Foundation recommended levels for young adults and adults of 7 to 9 hours of sleep per night.[1] The mean total sleep time in the present study was indeed <7 h/night with the 50% of the study sample (8/16 subjects) displaying always insufficient sleep time from PRE2 to POST1. In addition, we observed that total sleep time did not vary across the study nights. The latter result is in contrast with previous data showing shorter total sleep time in athletes the night of a competition compared with surrounding nights, and it seems even more challenging for players to obtain optimal sleep after evening matches.[5,13] Fullagar et al[26] examined the sleep of top-level soccer players and observed a reduced perceived recovery and total sleep time following night matches compared with day matches (−157 min) and training days (−181 min). Similarly, shorter total sleep time was observed the night of night matches compared with training days in team sport athletes.[13,14] The reductions in total sleep time the night of an evening competition may be explained by a delay in bedtime, which may in turn reduce the overall time spent in bed. To confirm this, it has been shown that Australian Rules Football players initiated sleep +2.5 hours later and obtained less total sleep time the night immediately after the evening game than a day game.[27] In line with this trend, Nedelec et al[13] observed that total sleep time was reduced in elite soccer players following a night match, and this difference was explained by the delayed bedtime. In the present study, we observed a +1.5-hour delayed bedtime in the first night following the competition compared with the surrounding nights. However, overall total sleep time was not reduced because players also woke up significantly later in the morning after the night match (+85 min compared with PRE2 and POST1). The delayed bedtime may be attributed not only to postcompetition players' activities (ie, social meetings, recovery or medical interventions, meals, shower and/or travels) but also to the fact that evening competition interferes with the psychophysiological changes that could delay sleep onset (eg, increased arousal, core body temperature, cortisol production, and/or exercise-induced muscle soreness).[28] On the other hand, the morning following the night match represented an opportunity for the players to extend their sleep as no morning training session was scheduled. It has indeed been observed that athletes' sleep behavior may change depending on the time of training,[29] including a later bedtime and wake time when not required to attend a morning training.[30]

For what concerns sleep quality, we did not observe significant changes across the nights in any of the sleep parameters. Importantly, independently of bedtime shifts, sleep stages maintained a good stability across the night as highlighted by slow wave sleep (N3), the deepest, and most restorative sleep stage,[31] which displayed mean percentage values between 24.3 (11.7) (EM) and 25.2 (10.1) (PRE2) across the nights. In addition, WASO and cortical arousals indices were stable as confirmed also by SE. This is of particular relevance, since it seems that the variation observed in bedtime did not have a detrimental effect on sleep, exhibiting a certain degree of adaptability among these athletes. In addition, the TQR and subjective sleep quality scores showed the same trend. Nevertheless, a recent meta-analysis by Roberts et al[5] highlighted that athletes rarely achieved SE recommendations on the night of competition, and the authors concluded that SE is reduced the night of competition compared with precompetition nights, especially

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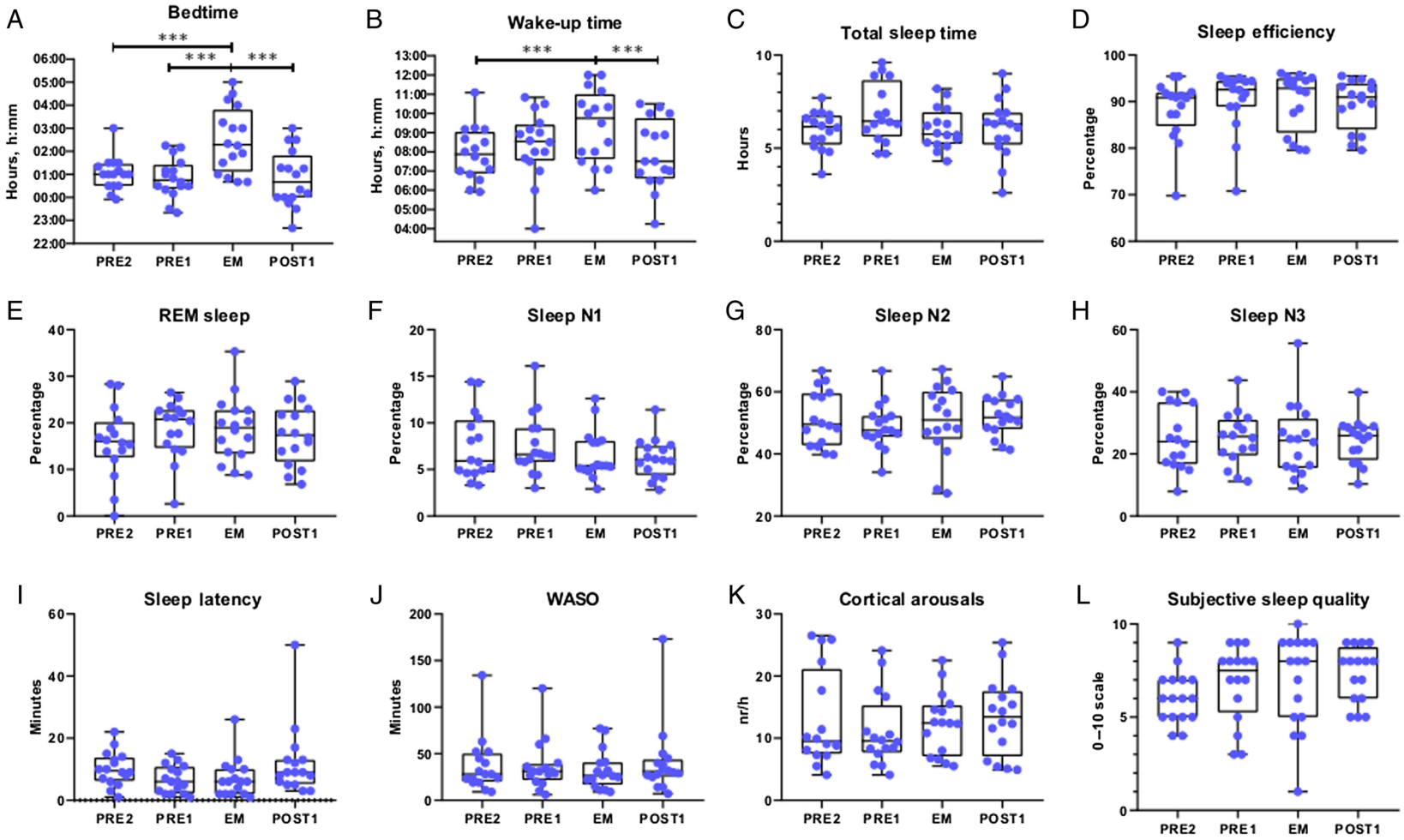

Figure 1 — Box-and-whisker plot with single data (circles) median, first, and third quartiles and minimum and maximum values of objective (A–M) and subjective (N) sleep parameters collected 2 nights before (PRE2 and PRE1) and 2 nights after an evening match (EM and POST1). nr/h indicates number of cortical arousals per hour of sleep time. ***$P < .001$.

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Table 1 Sleep Parameters and TQR Values 2 Nights Before (PRE2 and PRE1) and 2 Nights After a Late-Evening Match (EM and POST1) at 6:00 PM to 9:00 PM

	PRE2	PRE1	EM	POST1	RM ANOVA	F (df) η_p^2	Bonferroni post hoc test
Bedtime, h:min	01:01 (00:25)	00:50 (00:31)	02:29 (01:15)	00:49 (01:08)	$P < .0001$	$F_{3,15} = 10.62$; $\eta_p^2 = .68$, large	EM > PRE2 ($P = .0007$); EM > PRE1 ($P = .0001$); EM > POST1 ($P = .0001$)
Get-up time, h:min	07:54 (01:22)	08:21 (01:45)	09:20 (01:55)	07:54 (01:49)	$P = .033$	$F_{3,15} = 3.17$; $\eta_p^2 = .39$, large	EM > PRE2 ($P = .050$); EM > POST1 ($P = .049$)
Total sleep time, h	5.9 (1.0)	6.8 (1.6)	6.0 (1.1)	6.1 (1.6)	$P = .253$	—	—
SE,[a] %	88.3 (6.4)	90.2 (6.6)	90.0 (6.2)	89.4 (5.2)	$P = .629$	—	—
REM sleep, %	15.8 (7.6)	18.6 (5.1)	18.6 (6.9)	17.3 (6.5)	$P = .461$	—	—
N1,[a] %	7.5 (3.6)	7.7 (3.2)	6.5 (2.6)	6.2 (2.1)	$P = .373$	—	—
N2, %	51.5 (8.9)	48.9 (7.3)	50.6 (11.5)	52.1 (6.5)	$P = .497$	—	—
N3, %	25.2 (10.1)	24.8 (8.5)	24.3 (11.7)	24.4 (7.1)	$P = .926$	—	—
Sleep-onset latency,[a] min	10.1 (5.4)	6.7 (4.4)	7.1 (6.2)	12.0 (11.4)	$P = .095$	—	—
WASO,[a] min	37.5 (30.0)	36.7 (27.0)	32.6 (21.0)	40.9 (38.2)	$P = .834$	—	—
Cortical arousals,[a] min	13.0 (7.8)	11.2 (5.9)	12.3 (5.0)	13.3 (6.2)	$P = .430$	—	—
Subjective SQ,[a] 0–10 scale	6.0 (1.3)	6.8 (2.0)	6.9 (2.5)	7.3 (1.4)	$P = .254$	—	—
TQR (6–20 scale)	12.5 (1.7)	14.0 (3.0)	13.1 (3.6)	14.9 (2.2)	$P = .141$	—	—

Abbreviations: df, degrees of freedom; REM, rapid eye movement; RM ANOVA, repeated-measures analysis of variance; SE, sleep efficiency; SQ, sleep quality; TQR, total quality of recovery; WASO, wake after sleep onset. Note: The data are reported as mean (SD). The variables were subjected to the RM-ANOVA followed by Bonferroni post hoc test or to the Friedman test followed by Dunn multiple comparisons for nonnormally distributed variables.

[a]Nonnormally distributed variables.

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following a night competition. Many environmental and physiological factors are able to explain the negative impact of the night competition on athletes' sleep quality: the higher light exposure inside the stadiums in the night hours, the increased postexercise circulating cortisol and body temperature, the increased sympathetic hyperactivity and muscle pain.[32] To confirm this, Dunican et al[33] reported lower values of SE the night of a match compared with surrounding nights in rugby players, and similar results were observed in netball players: Sleep quality following evening competition was reduced compared with the nights of rest days.[34] A key variable that could partially explain the discrepancies in the results on SE in response to a night competition may be the time difference between the end of competition and bedtime. A delayed bedtime following a night competition may paradoxically represent a protective factor for athletes' SE. It is indeed reasonable to suppose that suggesting an athlete go to sleep immediately after a night match (or close to the end time of competition) could increase the possibility that the player has a disrupted sleep with longer sleep onset latency and lower SE, as already observed in nonathlete populations.[35] In line with this interpretation, in the present study, we observed a delayed bedtime (with no difference in sleep time due to a concomitant delay in wake-up time) with no deterioration of sleep quality parameters. Importantly, a striking difference between our study and those previously published[34,35] is that we used a portable PSG and not actigraphy to objectively quantify sleep.

A novelty of the present study is the use of a sleep device that has been shown to provide valid information on sleep architecture and continuity biomarkers when compared with PSG.[17] Although PSG is the gold standard method to measure sleep architecture, the participants are always asked to sleep in an unfamiliar environment, such as a sleep laboratory or hospital, for several nights, which could alter sleep patterns. Therefore, the use of PSG to study athletes' sleep is not always possible. On the contrary, a portable polysomnographic device may represent a valid option that is able to provide more detailed information than actigraphy to assess athletes' sleep architecture when sleeping at home without interfering with their training, competition, and sleep routines.

Some limitations need to be acknowledged. First, we studied athletes' sleep in response to a single night competition and for a short timeframe (ie, 4 consecutive nights), and longer periods of sleep monitoring along the entire competitive season are needed to confirm the present results. Sleep characteristics during competition period should indeed be compared with habitual sleep values from an appropriate baseline period. Second, although the device provided valid information on athletes' sleep characteristics, the players reported a feeling of intolerability and some difficulties in wearing the device due to its shape and size that mainly limited the possibility to sleep while lying facedown. As a consequence, it is unlikely to utilize this device for longer periods. Third, even if we reached the adequate power, the sample size used was low, limiting the significance of the results. Fourth, the players of this study were recruited from 3 different teams (ie, basketball, field hockey, and volleyball), with different competition levels, and the results may not be generalizable to other sport disciplines and to all the competition levels; in addition, no interteams and interchronotypes differences were investigated due to the small sample size of each subgroup. Fifth, athletes' napping behavior was not evaluated; daytime nap could possibly impact the nighttime sleep, and it also represents an advantageous intervention to enhance recovery process in athletes.[36,37] Finally, this was a field-based study; therefore, some factors, other than the timing of competition (eg, use of caffeine, alcohol, and/or electronic devices before bedtime), have not been controlled and may have affected the players' sleep/wake behavior.

Practical Applications

Team sport players have delayed bedtime and wake-up time following an evening competition with no negative consequences for sleep quality parameters. These results suggest that forcing players to go to bed early after an evening match could be counterproductive as a short time period between the end of the competition and bedtime could potentially lead to sleep disturbances. In addition, the morning after a night match should be free of training sessions or social commitments so that athletes may have the opportunity to extend their sleep time.

Conclusions

Our results suggest that team sport players have delayed bedtime and wake-up time following an evening competition; however, sleep architecture, duration, and subjective scores of recovery were not affected by a night match.

Acknowledgments

This study was supported by the Italian Ministry of Health (Ricerca Corrente). Mr D.J.L. is an employee of and shareholder in Advanced Brain Monitoring, Inc. He would benefit financially if ownership of the Sleep Profiler intellectual property were to be sold to a third party.

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